

AI-Based Smart Helmet Detection System

This project is **different from the previous smart helmet projects** you shared.

The previous projects focused on **building a smart helmet with sensors (IoT)**.

This project focuses on **detecting whether riders are wearing helmets using Artificial Intelligence and Computer Vision** through CCTV cameras or traffic cameras. It does **not require any electronics or sensors inside the helmet**. Instead, it automatically detects helmet violations from images or live video.

What Problem Does This Project Solve?

The project addresses several real-world problems related to road safety and traffic enforcement.

1. Riders Not Wearing Helmets

Many motorcycle riders ignore helmet rules, increasing the risk of severe head injuries during accidents.

Solution

The AI system automatically detects whether a rider is wearing a helmet.

2. Manual Traffic Monitoring

Traffic police cannot continuously monitor every road.

Solution

The system automatically monitors traffic using cameras without requiring human supervision.

3. Difficulty Identifying Traffic Violators

After detecting a rider without a helmet, authorities still need to identify the vehicle.

Solution

The system captures the rider's image and extracts the vehicle's registration number using OCR.

4. Slow Law Enforcement

Manual checking consumes time and manpower.

Solution

The AI system performs automatic detection and records violations in real time.

5. Lack of Safety Data

Authorities have limited information about helmet compliance.

Solution

All detected violations are stored in a database for analysis and future reporting.

These problems and objectives are described in the introduction and proposed system sections.

Main Idea of the Project

The project uses **Artificial Intelligence** and **Computer Vision** to analyze images and live video streams.

The system works as follows:

1. A CCTV camera captures live traffic.
2. The AI model detects motorcycles and riders.
3. It determines whether each rider is wearing a helmet.
4. If a rider is not wearing a helmet:
 - The system captures an image.
 - OCR reads the vehicle's number plate.
 - The violation is stored in the MongoDB database.
5. Authorities can later use this information for enforcement.

Unlike IoT-based helmets, this project works entirely through cameras and AI software.

Features of the Project

1. Real-Time Helmet Detection

Problem

Traffic police cannot continuously monitor all roads.

Solution

The AI model continuously analyzes live video from traffic cameras.

Whenever a rider appears:

- Helmet detected → Safe
- Helmet not detected → Violation recorded

Benefit

- Automatic traffic monitoring
 - Reduces manual work
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2. AI-Based Object Detection

The system uses **YOLO (You Only Look Once)**, a deep learning object detection algorithm.

YOLO detects:

- Person
- Motorcycle
- Helmet
- License plate

It works in real time and is suitable for traffic surveillance.

3. Helmet Classification

After detecting a rider, the AI classifies whether the rider is:

- Wearing Helmet 
- Not Wearing Helmet 

The result is displayed immediately on the screen.

Example screenshots are shown on **pages 3 and 4**, where detected riders are highlighted with bounding boxes and labels.

4. Number Plate Recognition (OCR)

Problem

Authorities need the vehicle number to issue fines.

Solution

The system uses **Optical Character Recognition (OCR)**.

OCR:

- Reads the vehicle number plate.
- Converts it into editable text.

Benefit

- Automatic identification of violators.
 - Faster fine generation.
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5. Database Storage

Whenever a violation occurs:

The system stores:

- Rider image
- Helmet detection result

- Vehicle number
- Other related information

using **MongoDB**.

Benefit

- Easy record management.
 - Future analysis.
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6. Image Detection

The application allows users to upload a single image.

The AI detects:

- Helmet
- No Helmet

The interface before and after detection is shown on **pages 3 and 4**.

7. Video Detection

The system also supports video files.

Users upload a video.

The AI processes every frame and detects riders wearing or not wearing helmets.

Screenshots of the video upload interface and detection output are shown on **pages 4 and 5**.

8. Traffic Monitoring

The software continuously monitors:

- Motorcycles
- Riders
- Helmet compliance

This makes it useful for:

- Smart cities
 - Highways
 - Traffic signals
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Technology Used

Technology	Purpose
Python	Backend programming
YOLO	Real-time object detection
CNN	Helmet image classification
OCR	Number plate recognition
OpenCV	Image and video processing
MongoDB	Store violation records
CCTV Camera	Capture live traffic
AI / Deep Learning	Automatic helmet detection

These technologies are described in the literature review and proposed system sections.

System Workflow

Traffic Camera



Capture Live Video



YOLO Detects Rider



Helmet Detection



Helmet?



YES



NO



Ignore



Capture Image



OCR Reads Number Plate



Store in MongoDB



Display Violation

Applications

According to the paper, this system can be used in:

Traffic Enforcement

Automatically detect helmet violations.

Road Safety Campaigns

Analyze helmet usage and identify high-risk areas.

Insurance Assessment

Support insurance companies in assessing rider safety.

Fleet Management

Monitor helmet compliance for company riders.

Accident Investigation

Provide evidence showing whether the rider was wearing a helmet.

Advantages

- Fully automatic helmet detection.
- No smart helmet hardware required.
- Real-time traffic monitoring.
- Detects helmet violations accurately.
- Reads vehicle number plates automatically.
- Stores records digitally.
- Reduces manual traffic policing.
- Supports large-scale deployment.
- Improves road safety compliance.

These benefits align with the paper's objectives and conclusion.

Limitations

The paper does not provide a separate limitations section. Based on the described system:

- Detection accuracy depends on camera quality.
 - Poor lighting or bad weather may reduce performance.
 - Occluded or blurred number plates may affect OCR accuracy.
 - High-performance hardware may be needed for real-time processing.
 - Detection performance depends on the quality of the trained AI model.
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Future Scope

The paper proposes several future enhancements:

- More advanced deep learning models for better accuracy.
 - IoT integration for remote monitoring and real-time analytics.
 - Wearable devices that alert users about helmet compliance.
 - Data analytics for predicting accident-prone areas.
 - Automatic compliance reports for regulatory authorities.
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Final Summary

This project is an **AI-powered Smart Helmet Detection System** that uses **computer vision and deep learning** to automatically identify motorcycle riders who are not wearing helmets. Unlike sensor-based smart helmets, it works with CCTV or traffic cameras, making it suitable for city-wide deployment. The system detects riders in real time, classifies helmet usage using **YOLO**, extracts vehicle registration numbers with **OCR**, stores violation records in **MongoDB**, and supports automated traffic enforcement. By reducing manual monitoring and improving compliance with helmet regulations, it aims to enhance road safety and make traffic management more efficient.